

DATA- Data is [raw facts](#) stored or used by a computer.

INFORMATION- Information is [organized data](#) with meaning, used for decisions and actions.

DAT ABASE- an organized collection of data, it can be easily accessed and managed

You can **organize data into tables, rows, and columns and use indexes to find information easily.**

DATABASE MANAGEMENT SYSTEM- system software for creating and managing databases.

(Ex : [MySQL, PostgreSQL, Oracle, Microsoft SQL Server.](#))

Responsibilities of a DBMS:

- Data storage and retrieval
- Data security and authorization
- Backup and recovery
- Concurrency control
- Data integrity enforcement

RDBMS- store data in tables with rows and columns and use SQL for data manipulation. Library Management System, Hospital Management System,

(Data store in as a table. Ex : [MySql, OracleDB, MsAccess](#))

Features:

- [Structured schema](#)
- [Strong consistency](#)
- [Uses relationships between tables](#)

NRDBMS-Used to store, manage & retrieve data in non-relational databases

Document-based, Key-Value

(Data store in as a document.Ex : MongoDB, Apache Cassandra, Redis

Features:

- Schema-less or flexible schema
- High scalability
- Used in big data and real-time systems

Database models- define the logical structure & organization of data in a DBMS.

(It acts as a blueprint for storing, accessing, updating data, and maintaining data integrity.)

Why Data Models are Important?

- Provide a clear structure for data
- Help in database design and development
- Improve data consistency and integrity
- Act as a communication tool between developers and stakeholders

1.Conceptual Data Model

represents data at a high level, focusing on what is needed, not how it is stored, and is used by business analysts and stakeholders.

Characteristics

- Independent of DBMS
- Easy to understand
- Uses business concepts
- Created during requirement analysis

Components

- Entities
- Attributes
- Relationships

Example

- Entity: STUDENT, Attributes: student_id, name, email

2.Logical Data Model

the data structure in detail, independent of physical storage, acting as a bridge between conceptual and physical models.

Characteristics

- Defines tables, columns, data types
- Shows relationships and constraints
- Normalized to reduce redundancy

Components

- Tables
- Primary keys
- Foreign keys
- Constraints

Example

STUDENT(student_id PK, name, email)

Usage

- database designers . Acts as a bridge between conceptual and physical models

3.Physical Data Model

how data is actually stored in the database, including types, keys, indexes, and constraints.

Characteristics

- DBMS-specific

- Includes indexing and storage details
- Performance-oriented

Components

- Table structures
- Indexes
- Storage engines
- Partitioning

Example: creat table example

Usage

- DBAs and developers
- Optimized for performance

client-server - : Clients send requests, and the server responds with data.

(distributed computing model)

Database Clients

- MySQL Workbench
- DataGrip
- pgAdmin
- DBeaver
- Web applications

Database Servers

MySQL Server
PostgreSQL
Oracle Database
MongoDB (NoSQL)

Advantages of Client–Server Architecture

- Centralized data management
- Improved data security
- Supports multiple users
- Easier backup and recovery

Disadvantages of Client–Server

- Server failure affects all clients
- Network dependency
- Higher setup and maintenance cost
- Requires skilled administration

- Better performance than file-based systems

How Client–Server Communication Works

Client and server communicate using the TCP/IP network protocol

Steps:

1. Client connects to the database server
2. Client sends SQL query/request
3. Server processes the query
4. Server retrieves or modifies data
5. Server sends results back to the client

Database as a server- runs a DBMS to centralize storage, security, access for multiple users.

server what to do (Listens for client requests, Executes SQL queries, Returns results to clients, Manages concurrent users)

SQL- language

MYSQL- a software

What is the Relational Model? the most widely used data model in dbms

Statement Categories

- DDL – Data Definition Language - Create, Alter, Drop (**define and modify database structures.**)
- DML – Data Manipulation Language - Insert, Update, Delete (**manage data inside tables.**)
- DQL – Data Query Language - Select (**retrieve data from tables.**)

Common Data Types

- INT – integer values
- VARCHAR(n) – text data
- DATE – date values
- DECIMAL(p,s) – decimal numbers

Constraints

Constraints are rules applied to columns to maintain **data integrity**.

- NOT NULL - no empty value
- UNIQUE - no duplicate
- PRIMARY KEY - unique and not null
- FOREIGN KEY - links tables
- DEFAULT - auto value if missing
- CHECK - conditions, based validation

Type of Attributes

simple - cant be divided

single valued - can be split into parts

multi valued -multiples values

derived - calculated from other attributes

stored - physical stored

key Attributes - uniquely identifies an entity

QUERY PART

- `mysql -u root -p`
- `docker ps;`
- `docker exec -it mysql mysql -h localhost --port 3306 -u root -p( UR PASSWORD );`
- `CREATE DATABASE ( database name );`
- `SHOW DATABASES;`
- `USE ( database name );`
- `CREATE TABLE ( table name );`
 - `id INT AUTO_INCREMENT PRIMARY KEY,`
 - `name VARCHAR(50) NOT NULL,;`
 - `email VARCHAR(100) UNIQUE,`
 - `gender ENUM('Male', 'Female', 'Other') NOT NULL,`
 - `birthday DATE NOT NULL,`
 - `payment DECIMAL(10,2) NOT NULL,`
 - `salary INT NOT NULL CHECK (salary > 30000),`
 - `salary DECIMAL(10,2) NOT NULL DEFAULT 35000`
 - `salary DECIMAL(10,2) NOT NULL DEFAULT 35000 CHECK (salary > 30000)`
 - `department_id INT, FOREIGN KEY (department_id) REFERENCES department(department_id)`
- `SHOW TABLES;`
- `DESCRIBE ( table name );` (tables details)
- `INSERT INTO ( databse name VALUES (int, 'string');`
- `SELECT * FROM ( table name );`
- `DELETE FROM gem WHERE id = 1;`

DROP TABLE (table name);

- DROP TABLE (table name);
- DROP DATABASE (database name);
- EXIT;De